LOOMBENCH BRIEF · GROUP COMPOSITION · 12 OCTOBER 2026

# Substituting one agent's disposition in a homogeneous group of five LLM agents

**Question.** When five agents given the same disposition divide a shared resource, how does the final allocation differ if one of them is given another disposition instead?

**Design.** Five agents each start with 20% of a resource and act for 26 rounds: maintain, claim a neighbour's part, concede, or propose joint ownership; the neighbour accepts, refuses or counters. Each agent's prompt holds a one-sentence disposition and no objective. At k = 0 all five agents have disposition A; at k = 1 one of them has disposition B, and so on to k = 5. Runs are paired by seed: a match number that gives the same starting set-up (seating, starting designs and random draws). The substituted position rotates over seeds. 10 seeds for k = 0 and 1, 5 for k = 2 to 5; each run executed once. Decision systems: Jev, GPT (Decisions), GPT (chat).

**Result.** In 6 comparisons the focal agent's mean final share was 38.8% to 46.6% at k = 1 against 9.1% to 19.6% at k = 0, and the other agents' mean share was lower. Systems whose change in the focal agent's share was not clearly different still differed in the claims and refusals recorded.

**Focal agent**: the agent whose disposition is substituted. **Other agents**: the four whose disposition is unchanged, reported as their mean.
**Share**: the percentage of the resource an agent holds alone at the end; a part under joint ownership counts for neither holder. **Seed**: a match number, used to pair runs.
**k**: the number of agents with the second disposition. **Intervals**: Student-t 95% over seeds.
**Systems**: Jev is jev-1.13.0; GPT (Decisions) is gpt-6-luna through OpenAI's Decisions API; GPT (chat) is gpt-6-luna as a chat model that states a probability for each option.

**calm** "Jessica is calm and relaxed, hard to upset, and keeps an even mood under stress."

**blunt** "Jessica is blunt and hard-headed, suspicious of people's motives, and does not mind starting an argument."

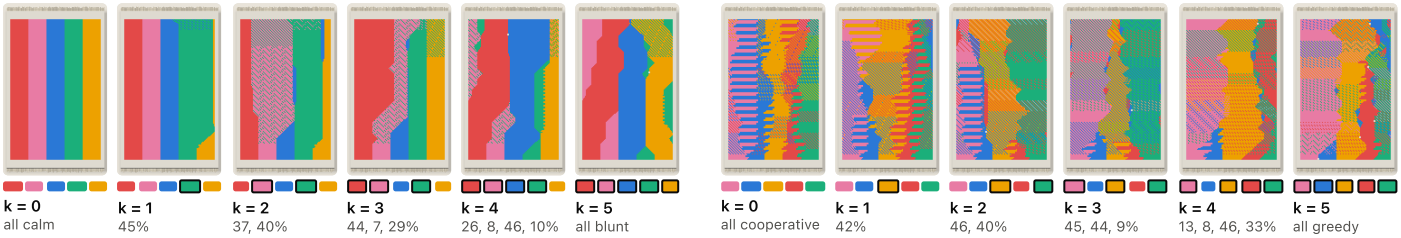
**rude** "Jessica is rude and inconsiderate, talks over people and talks down to them, and does not care if she gives offence."

**cooperative** "Jessica is cooperative and fair-minded, likes to work with others toward what suits everyone, and is glad to share."

**greedy** "Jessica is greedy and self-interested, wants more than her share, and takes what she can get."

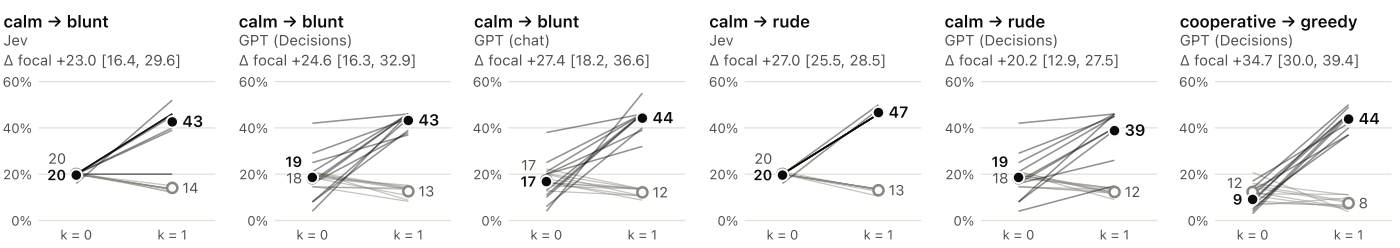
## FIGURE 1 Final allocation in one seed as k agents are given the second disposition

Each picture shows who held which part of the resource in each round (time runs bottom to top), one colour for each agent; hatching marks a design adopted from another agent, two-colour bands a part under joint ownership. Squares give the seating, outlined for agents with the second disposition, whose final shares are beneath (k = 1 to 4). The seed shown is the one closest to the mean result at k = 1.




## FIGURE 2 Final share at k = 0 and k = 1, by seed

Each line is one seed (10 in each panel); dots and numbers are means. Δ focal is the mean paired change in the focal agent's share, in percentage points, with its 95% interval.




## FIGURE 3 The same contrast under each decision system

Means with 95% intervals over seeds. Points and intervals share a scale within each column, and the vertical line marks zero. The last rows give paired differences between systems with their intervals; bold where the interval excludes zero.

| calm → blunt | Focal agent: change in final share (pp) | Other agents: change in final share (pp) | Focal agent: claims per match, k = 1 | Other agents: refusals per agent per match, k = 1 | All agents: claims per agent per match, k = 0 | Resource under joint ownership, k = 0 (%) |
|---|---|---|---|---|---|---|
| **Jev** | +23.0 | −6.0 | 4.3 | 0.0 | 0.0 | 0.0 |
| **GPT (Decisions)** | +24.6 | −5.7 | 10.3 | 0.8 | 1.5 | 8.4 |
| **GPT (chat)** | +27.4 | −5.3 | 7.5 | 0.6 | 1.1 | 13.6 |
| GPT (Decisions) − Jev<br>paired, 10 seeds | +1.6 [-10.8, 14.0] | +0.4 [-2.9, 3.6] | +6.0 [3.9, 8.1] | +0.8 [0.5, 1.2] | +1.5 [1.1, 1.9] | +8.4 [3.6, 13.2] |
| GPT (chat) − Jev<br>paired, 10 seeds | +4.4 [-4.5, 13.3] | +0.8 [-1.4, 2.9] | +3.2 [1.5, 4.9] | +0.6 [0.3, 0.8] | +1.1 [0.8, 1.5] | +13.6 [6.4, 20.8] |

## Findings

1. **The focal agent's mean final share was higher at k = 1 in every comparison, and the other agents' lower.** Mean paired change +20.2 to +34.7 percentage points for the focal agent and −7.0 to −4.8 per other agent, across the 6 comparisons (10 seeds each; every 95% interval excludes zero; Figure 2).
2. **The paired change in the focal agent's share was not clearly different between Jev and GPT (Decisions); the focal agent's claims and the other agents' refusals were (calm → blunt; Figure 3).** Paired over 10 seeds, GPT (Decisions) minus Jev: change in the focal agent's share +1.6 [-10.8, 14.0] pp; claims by the focal agent 10.3 against 4.3 per match (+6.0 [3.9, 8.1]); refusals per other agent 0.8 against 0.0 (+0.8 [0.5, 1.2]). GPT (chat): change in the focal agent's share +27.4 [18.2, 36.6] pp.
3. **Homogeneous calm groups differed between systems.** At k = 0, Jev agents made 0.0 claims per agent per match and averaged 20.0% each; GPT (Decisions) agents made 1.5 (+1.5 [1.1, 1.9]), with 8.4% of the resource under joint ownership.
4. **Other disposition comparisons.** Calm → rude: the change in the focal agent's share was smaller under GPT (Decisions) than under Jev (−6.8 [-13.3, -0.3] pp, 10 seeds). Cooperative → greedy, GPT (Decisions): 41.6% of the resource was under joint ownership at k = 0 and 26.2% at k = 1 (−15.4 [-29.3, -1.5]).
5. **Mean share per substituted agent was lower at larger k.** Jev, blunt: 43, 38, 29, 23, 20%; GPT (Decisions), blunt: 43, 41, 32, 25, 20% for k = 1 to 5 (5 seeds from k = 2). At k = 4 the one remaining calm agent held 8.4% (Jev) and 1.6% (GPT (Decisions)).

## Limitations

Each run was executed once, so run-to-run variation is unmeasured; intervals describe variation over seeds. Each disposition is a single wording. A difference for the other agents is between the two resulting matches: it includes everything that followed from the substitution and does not isolate their direct response to the focal agent. A comparison between systems compares a model together with its interface and the way its probabilities are read, not models alone. The results concern this task and these systems.

Every match and table · `node analysis/mix.mjs` prints the numbers.